Developmental Cognitive Interpretability: paper report and relevance to Explore Persona Space

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1 What the work is

Two linked artifacts by Jason R. Brown (Cambridge) and Edward James Young (Geodesic Research):

1. **The agenda post**: "Developmental Cognitive Interpretability: A Research Agenda" (LessWrong, May 29, 2026). Proposes predicting out-of-distribution (OOD) AI behavior by modeling how interpretable cognitive constructs — values, goals, motivations — *evolve over training*, rather than interpreting the final model's internals or evaluating its behavior in snapshot.
2. **The companion paper**: "Understanding Goal Generalisation in Sequential Reinforcement Learning" (arXiv:2605.23565). The toy-scale demonstration: 298 CNN maze agents trained on 100+ sequential goal-pursuit pipelines, evaluated on 250+ OOD environments, with a 10-dimensional latent model ("latent policy gradients") that predicts each agent's full OOD preference profile *from the training pipeline description alone.*

2 Positioning and motivation

The agenda carves out a third level of analysis between the two standard approaches. Mechanistic interpretability works at Marr's implementation level (weights, activations, circuits); behavioral evaluation works at pure input–output. *Cognitive* interpretability sits at the computational/algorithmic level: explain behavior through latent constructs (goals, beliefs, preferences) without committing to how they are implemented in the network.

The safety motivation is **behavioral degeneracy**: identical behavior on the tested distribution is compatible with conflicting underlying cognition (genuine alignment vs. reward-seeking vs. scheming). Behavioral evals cannot distinguish these by construction; mechanistic interpretability could in principle but does not yet scale to the question. If the latent cognitive variables can be recovered, behavior on *untested* inputs becomes predictable — which is the actual deployment-safety question.

The **developmental** move is the distinctive part: model how the cognitive constructs change as a function of the training pipeline. The rationale is that post-training objectives underspecify behavior across domains, while earlier training stages shape the generalizable motivations; if each stage's effect on the latents can be modeled, stages can be *composed* to predict the outcome of a novel pipeline before it is run.

2.1 The four load-bearing assumptions

A1 — OOD behavior is structured. Behavior beyond the training distribution is non-arbitrary. Cited support: LLMs show identifiable values and systematic tendencies that scale predictably. Acknowledged counterevidence: LLM behavior is highly conditional and sensitive to spurious correlations. If false, nothing downstream works.

A2 — the structure compresses into interpretable latents. A small set of human-meaningful variables captures the behavioral structure. Demonstrated in the toy domain (276 choice probabilities $\rightarrow$ 24 value scores, §3); for LLMs, supported by low-dimensional internal representations of constructs such as personas. If false, prediction without interpretation.

A3 — latent evolution is predictable from training information alone. Architecture, optimizer, data, pipeline structure — *not* behavioral measurements of the trained model. The strongest and most distinctive claim. If false, development is too chaotic to forecast.

A4 — the maps generalize across pipelines. Both the training$\rightarrow$latents and latents$\rightarrow$behavior maps, learned on one family of models/pipelines, transfer to novel ones. Indirect support: neural scaling laws; alignment techniques working consistently across model sizes, families, and datasets. If false, every new pipeline needs its own empirical study and the approach has no predictive payoff.

The proposed workflow loop: probe behavior broadly (forced choice) $\rightarrow$ fit an interpretable cognitive model compressing the choices into latents $\rightarrow$ model how training moves the latents $\rightarrow$ compose stages to predict novel pipelines $\rightarrow$ validate on held-out pipelines.

3 The toy demonstration

Setting. CNN-based RL agents (4.8M parameters) navigate mazes toward goal objects defined by color $\times$ shape — 24 object types over 10 features. Training pipelines are *sequential*: e.g., stage 1 trains pursuit of black plusses, stage 2 retrains toward red circles; some stages include distractor objects. The behavioral probe is forced choice: place the agent in a maze with two object types and record which it reaches (or neither, within the 200-step horizon). With 24 object types this gives $\binom{24}{2} = 276$ pairwise probes per agent, almost all OOD — the agent only ever saw its specific trained goals.

Descriptive latents (the A2 test). Each agent's behavior is fit with the Elo algorithm (maximum-likelihood Bradley–Terry): each object gets a scalar score V, and $P(A \succ B) = \sigma(V_A - V_B)$ (logistic in the score difference). The three-way outcome (A / B / neither) is handled by entering "no goal" as a competitor and shifting scores so it sits at zero, making scores comparable across agents and reflective of absolute pursuit propensity. Boltzmann-rationality is a stochastic transitivity requirement, so the fit is falsifiable: intransitive or context-dependent preferences would not compress into 24 scores. Validation is 4-fold cross-validation over the pairs: scores fit on 3/4 of pairs predict the held-out quarter's preference direction with **89.73% accuracy** ($\pm$0.18% SE). Nothing in the training objective forces this coherence; it is an empirical finding about OOD preferences.

Empirical findings. Three patterns, each later reproduced by the model in §4:

1. **Feature salience is uneven.** Shape drives generalization more than color (plus-shape most salient, black least), consistent with known CNN inductive biases; training on one feature can raise or lower the value of others (*feature confusion*).
2. **Values persist across stages.** Goals learned in stage 1 retain value after stage 2, whether or not the goals share features; agents trained on more unique features generalize to pursue more objects.
3. **Repeated features strengthen; existing values inhibit new ones.** A feature shared across stages ends up stronger than if trained once; the non-shared feature of the second goal is valued much less than it would be alone — a gradient-starvation / plasticity-loss signature. Corollary: strong ordering effects when goals share a feature.

4 The latent policy gradient model

The core methodological contribution. Rather than interpreting the trained network, the model is a small, interpretable *simulation of the training process*: a 10-dimensional state whose values are written by gradient ascent, stage by stage, in the same way RL writes values into the real network.

State and readout. The latent state is $\boldsymbol{w} \in \mathbb{R}^{10}$ (one dimension per feature, not per object). An object with binary feature vector $\boldsymbol{\phi}$ has value $v = \boldsymbol{\phi} \cdot S\boldsymbol{w}$, and preferences are Boltzmann-rational in the values:

$$\Pi\Big(\boldsymbol{\phi}^{(g)} \,\Big|\, \boldsymbol{\phi}^{(g)}, \mathbf{0};\, \boldsymbol{w}\Big) = \frac{\exp\big(\boldsymbol{\phi}^{(g)} \cdot S\boldsymbol{w}\big)}{1 + \exp\big(\boldsymbol{\phi}^{(g)} \cdot S\boldsymbol{w}\big)}, \tag{1}$$

equivalent to modeling the Elo scores as linear in features. S is a 10×10 upper-triangular *saliency matrix* (upper-triangular to fix the rotational gauge freedom in $\boldsymbol{w}$).

Dynamics. To predict an agent trained on pipeline Ω, initialize $\boldsymbol{w} = w_0\mathbf{1}$ and, for each stage with goal features $\boldsymbol{\phi}^{(g)}$, run $\sim$100 steps of gradient ascent *on* $\boldsymbol{w}$ maximizing a modified policy objective

$$J = \Pi\big(\text{goal chosen}\big) + \tau\, h\big(\Pi(\text{goal chosen})\big), \tag{2}$$

the probability of pursuing that stage's goal plus a binary-entropy bonus. Each stage's final $\boldsymbol{w}$ initializes the next. The entropy term does real work: without it, "training" would push the goal's value to infinity. With it the gradient has the closed form

$$\nabla_{\boldsymbol{w}} J = \big(1 - \tau v^g\big)\, \sigma(v^g)\big(1 - \sigma(v^g)\big)\, S^T \boldsymbol{\phi}^{(g)}, \tag{3}$$

which vanishes at $v^g = \tau^{-1}$: training drives the goal's value to a finite ceiling and stops. The single scalar τ encodes how completely agents master their training objective.

Training is iterated projection. Because the gradient always points along the fixed direction $S^T\boldsymbol{\phi}^{(g)}$, each stage has an analytic solution: it projects $\boldsymbol{w}$ onto the hyperplane $\boldsymbol{\phi}^{(g)} \cdot S\boldsymbol{w} = \tau^{-1}$,

$$\boldsymbol{w}_{\text{new}} = \boldsymbol{w}_0 + \frac{\tau^{-1} - \boldsymbol{\phi}^{(g)} \cdot S\boldsymbol{w}_0}{\big\|S^T\boldsymbol{\phi}^{(g)}\big\|_2^2}\, S^T\boldsymbol{\phi}^{(g)}, \tag{4}$$

so a multi-stage pipeline is a sequence of projections. This single picture reproduces all three empirical findings:

- *Persistence*: each projection moves $\boldsymbol{w}$ along one direction only; what earlier stages wrote into orthogonal directions is untouched.
- *Inhibition*: the update magnitude is proportional to the value gap $\tau^{-1} - \boldsymbol{\phi}^{(g)} \cdot S\boldsymbol{w}_0$; if the new goal shares a feature with an old one, the gap is already small, so little new value is written — gradient starvation from a one-line mechanism.
- *Generalization as a similarity metric*: from $\boldsymbol{w}_0 = 0$, training on goal g changes another object $\boldsymbol{\phi}'$'s value by $\tau^{-1}\, \boldsymbol{\phi}' \cdot SS^T \boldsymbol{\phi}^{(g)} / \|S^T \boldsymbol{\phi}^{(g)}\|_2^2$. The matrix SS^T is therefore an *object similarity metric as judged by the architecture*: training on one thing raises the value of everything else in proportion to similarity under this metric. Its diagonal entries are per-feature learning rates (salience); off-diagonals are cross-feature bleed.

Fitting. Per-agent latents are never fit to that agent's behavior — they are generated from the pipeline description. The only fitted quantities are the shared hyperparameters (S, τ, w_0), 57 numbers across all 298 agents, optimized by minimizing the KL divergence between predicted and observed three-way choice distributions over ∼82k datapoints (298 agents × 276 probes), with gradients backpropagated *through the 100-step inner training simulation* (Adam, lr 0.03, batch 64, one epoch).

Evaluation. Four evaluations (full fit; 4-fold CV holding out 25% of pipelines; transfer from single-stage to two-stage pipelines; transfer from no-distractor to distractor pipelines) against ablations that each delete one mechanism:

Variant	Deletes	Outcome
Diagonal S	feature confusion	worse — cross-feature bleed is real
Memoryless (last stage only)	training history	clearly worse — training is not memoryless
Simultaneous (ignore order)	stage ordering	close — ordering effects real but small
Quadratic features	parsimony	no gain

Two lower bounds (per-agent per-feature and per-agent per-goal value fits — the best any linear-in-features or any Boltzmann-rational model could do, with vastly more parameters) calibrate how much headroom remains. The proposed model achieves the lowest modeling loss in all four evaluations; the two transfer results — hyperparameters fit only on single-stage pipelines predict two-stage agents, and fit without distractors predict distractor pipelines — are the paper's direct evidence for A4 at toy scale. Model ranking is stable under total-variation distance, Brier score, and directional accuracy.

5 Scaling to LLMs and open problems

For LLMs the authors argue A1/A2 already have direct support (identifiable LLM values fit by Boltzmann-rational and Thurstonian models; low-dimensional internal persona representations, citing Anthropic's Assistant-Axis work), while **A3/A4 are undemonstrated** — only the indirect scaling-laws and cross-model-consistency evidence. They are candid that whether the agenda scales to LLMs at all is the primary uncertainty.

Six open problems: (1) scale latent policy gradients to simple character-training-style pipelines, reading out forced-choice value rankings and recovering personas as correlation structure among trait optimizations; (2) stress-test the toy setting across architectures and algorithms; (3) broaden the space of cognitive models before reaching for complex ones; (4) formalize informal theories (Persona Selection Model, Behavioural Selection Model, Shard Theory); (5) extend past RL to RLHF, DPO, deliberative alignment, prompt distillation, synthetic-document finetuning, and AI debate; (6) eventually model frontier training pipelines. The post closes with a collaboration call (EAG London 2026).

6 Relation to Explore Persona Space

This agenda is unusually close to the project — closer than most interpretability agendas — because the project is, in effect, already running the LLM-scale version of it.

1. **A3 is the pre-training prediction thread.** "Latent evolution predicts from training information alone" is the formal statement of the project's behavior-leakage-prediction question: predict emergent-misalignment and leakage outcomes from training-data signals *before* training. The persona-distance predictors (#404/#458: base-model KL/JS-divergence and cosine-similarity metrics predicting leakage) test a claim *stronger* than A3 — endpoint prediction from base model + data alone, with no dynamics fitting at all.

2. SS^T **is the leakage gradient.** The model's central interpretable object — a learned similarity metric under which training on one goal raises the value of others — is structurally the same hypothesis as the project's distance→leakage gradient over personas, and as the rank-1 leakage model (`docs/notes/rank1_leakage_model.pdf`): a constant write, gated by context similarity. Their toy result is evidence that such a metric (i) exists, (ii) is learnable from few pipelines, and (iii) transfers to unseen pipeline types.

3. **The project already has the developmental data the agenda needs.** Marker log-probability trajectories logged per training step per condition, the dose–response cliff at 10–25 EM-training steps (#104/#139/#125), and the band-stop training recipe are LLM-scale measurements of "latent evolution over training" — the quantity their method simulates. Their value-persistence and inhibition findings also rhyme with project phenomena: two-phase training (persona coupling, then EM induction) is a two-stage pipeline in their vocabulary, and contrastive negatives are explicit value-suppression at the slot.

4. **Personas are their proposed latents.** "Recover a notion of personas in the form of how optimising for certain character traits and values are correlated with one another" is close to a description of the project's axis-characterization line, and the sibling papers (Persona Vectors; Persona Features Control Emergent Misalignment) are concrete implementations of the latent variables they gesture at. Their motivating ambiguity — same behavior, different cognition — matches the finding that EM is a villain-character persona rather than rogue-AI cognition.

5. **The project's comparative advantage is their missing piece.** They have no LLM results and no mechanism for prediction at scale; open-weights Qwen, training-data ablations, and weight-space probes are exactly the tooling their Section 4 needs.

Divergences. They model *trajectories* (fit dynamics, forecast forward); the project's predictors so far predict *endpoints* from pre-training signals. Their toy domain has a ground-truth enumerable goal space and behavior fully determined by values; LLM behavior is conditional, persona-gated, and measurement-regime-sensitive (saturation, on-policy vs. teacher-forced) — exactly the regime where their A1 counterevidence bites and where the project's measurement discipline is the harder-won asset.

Possible follow-up. A direct A3/A4 test at LLM scale using existing project data: fit a low-dimensional latent-dynamics model (persona-space position with a learned similarity metric, the SS^T analogue) to marker log-prob trajectories across training mixes, and predict the full leakage trajectory — or at least the endpoint leakage profile — for held-out mixes. The single-stage→two-stage transfer evaluation has a natural analogue: fit on single-behavior implants, predict two-phase (coupling → EM) pipelines.

7 Assessment

The agenda framing is clean and the toy result is real: coherent OOD preferences were not guaranteed, the 57-hyperparameter model beating per-mechanism ablations across transfer splits is a substantive result, and the iterated-projection / similarity-metric interpretation is genuinely elegant. The gap to LLMs is the whole game, and the authors say so. Everything that makes the toy tractable — enumerable goal space, two-stage pipelines, behavior fully determined by values, a probe (forced choice) that is trivially on-distribution — is absent at scale. The most valuable near-term exchange runs in both directions: their framework supplies a formal vocabulary (latent dynamics, composable stages, learned similarity metric) for results the project already has; the project supplies the LLM-scale developmental measurements their agenda currently lacks.

References

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